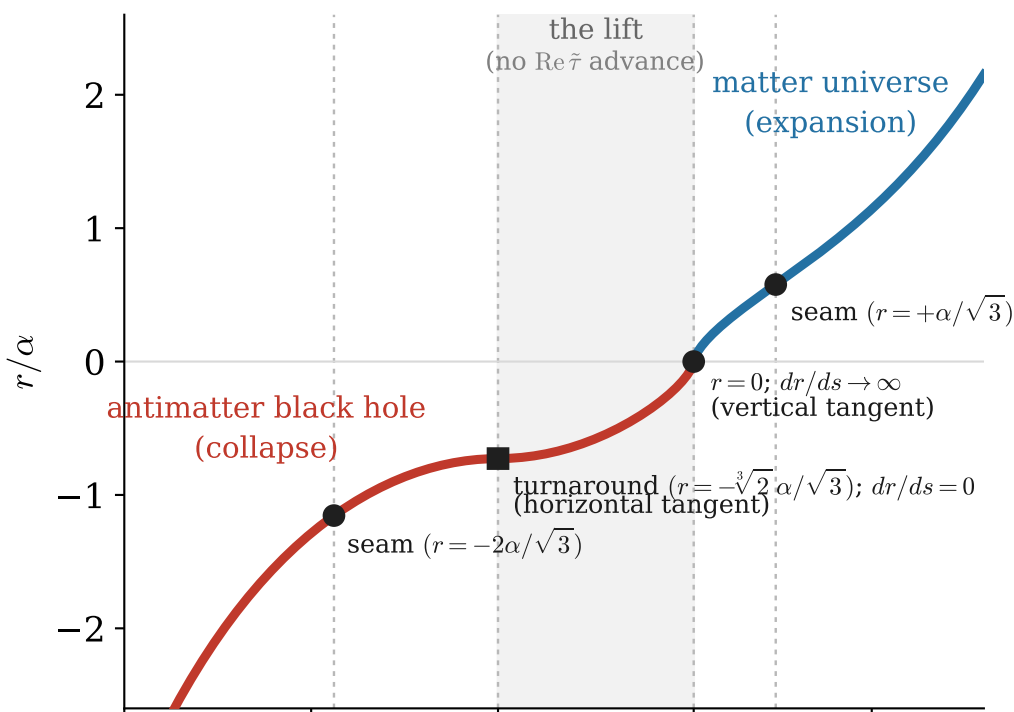
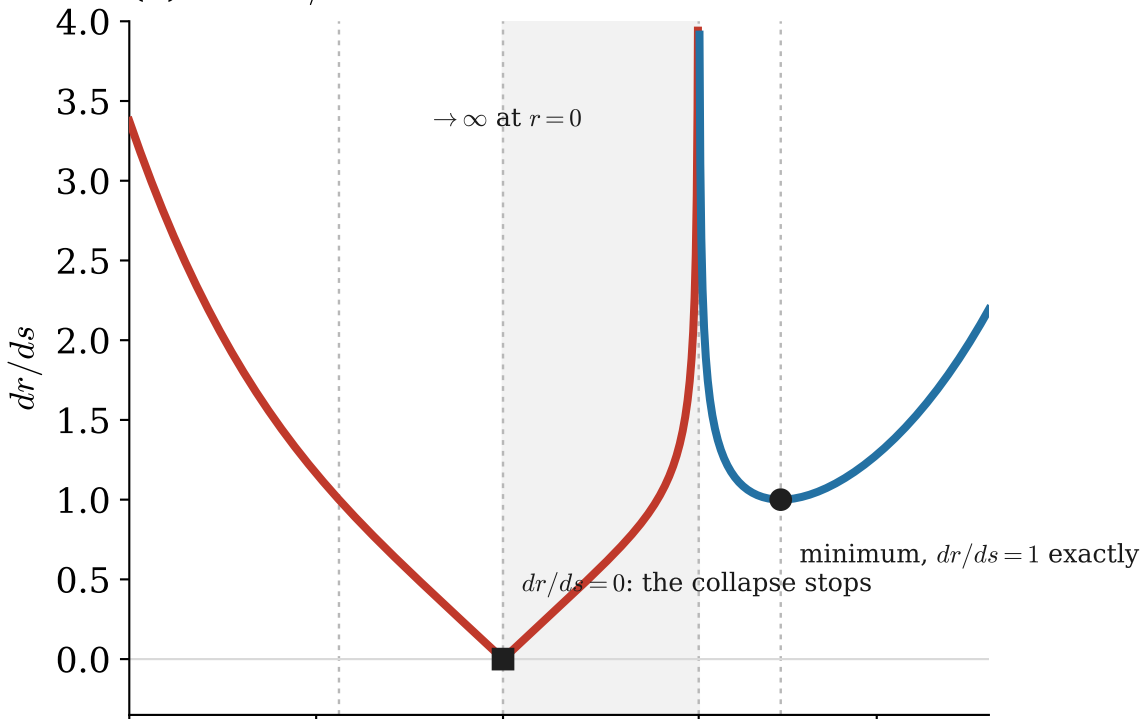


**(a) position  $r$**



**(b) rate  $dr/ds$**



**(c) acceleration  $d^2r/ds^2$**

